

Devops Lab Program 1: Build a container from custom docker file

Dockerfile

```
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg1
GNU nano 6.2 Dockerfile
# Use official Python image
FROM python:3.9-slim

# Set working directory inside container
WORKDIR /app

# Copy requirements first (Better for caching)
COPY requirements.txt requirements.txt
RUN pip install --no-cache-dir -r requirements.txt

# Copy all project files
COPY . .

# Expose the Flask default port
EXPOSE 5000

#Set Flask
ENV FLASK_APP=app.py
ENV FLASK_RUN_HOST=0.0.0.0
ENV FLASK_RUN_PORT=5000
# Run the Flask app
CMD ["python", "app.py"]
```

app.py

```
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg1
GNU nano 6.2 app.py
from flask import Flask
app = Flask(__name__)

@app.route("/")
def home():
    return "Hello from Flask in Docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

requirements.txt

```
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg1
GNU nano 6.2 requirements.txt
flask
```

```
1RV24MC075_Pragathi_h_r@dell: ~/DevOps/prg1
1RV24MC075_Pragathi_h_r@dell:~$ cd DevOps
1RV24MC075_Pragathi_h_r@dell:~/DevOps$ cd prg1
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg1$ nano Dockerfile
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg1$ nano app.py
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg1$ nano requirements.txt
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg1$ docker build -t flask-app .
[+] Building 2.6s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 490B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> [internal] load build context
=> => transferring context: 93B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt requirements.txt
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> CACHED [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:512d1f380e5fd4fc8ce65b91a1434ddf6d970afa56221609833306ea68a80afb
=> => naming to docker.io/library/flask-app
1RV24MC075_Pragathi_h_r@dell:~/DevOps/prg1$ docker run -p 5000:5000 flask-app
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
```

Output

